

DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Improving TA-LLM dialogue flow via automatic paired trajectory construction

A 5-state MDP formulation · 3 query types · Automatic preference pairs · Near GPT-4o performance

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs must decide: ask follow-up questions, make a tool call, or reject.

SFT with expert trajectories is insufficient to learn these conversational control skills.



Failure Mode 1

Missing Follow-Up Questions

Model skips slot-filling when arguments are incomplete



Failure Mode 2

Hallucinated Tool Invocations

Model fabricates arguments rather than asking the user



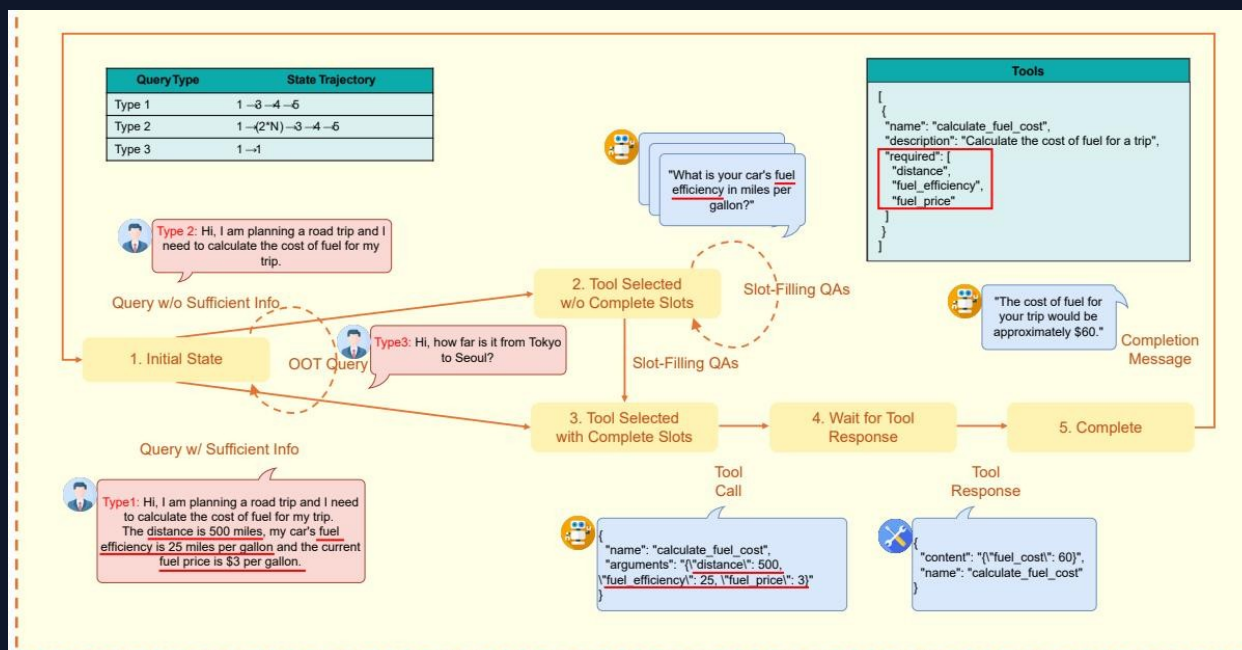
Failure Mode 3

Failed Tool Rejection

Model calls tools for out-of-scope requests instead of declining

→ DiaTool-DPO proposes Direct Preference Optimization to learn correct dialogue flow from automatic preference pairs

Modeling TA-LLM Dialogue as a 5-State MDP



S1

Initial

No history. Out-of-scope → reject & stay.

S2

Tool Selected

w/o Complete Slots

Tool identified, parameters missing. Slot-filling QA occurs here.

S3

Tool Selected

w/ Complete Slots

Tool selection and all argument values determined.

S4

Wait for

Tool Response

Tool call executed, awaiting results.

S5

Complete

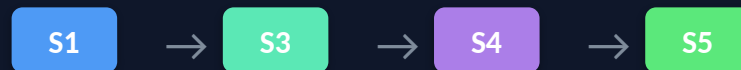
Tool results received, completion message generated.

Three Query Types Define Distinct State Trajectories

User queries are classified by state transition trajectory through the MDP:

Query Type	State Trajectory	Condition	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	All arguments present	"Book a flight from NYC to LA on Dec 25"
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Missing arguments (N rounds of slot-filling)	"Book a flight" $\rightarrow$ "Where from?" $\rightarrow$ "NYC" $\rightarrow$ "Where to?" $\rightarrow$ "LA" $\rightarrow$...
Type 3	$1 \rightarrow 1$	Out-of-scope request (no suitable tool)	"What's the meaning of life?" $\rightarrow$ Model rejects tool call

Type 1 (complete info)



Type 2 (incomplete info)



Type 3 (out-of-scope)



Automatic Paired Trajectory Construction Without Human Labels

Chosen (correct) vs. Rejected (incorrect) trajectories are paired automatically for each query type:

Query	Chosen Trajectory	Rejected Trajectory	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562	Easy/Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089/562	Easy/Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

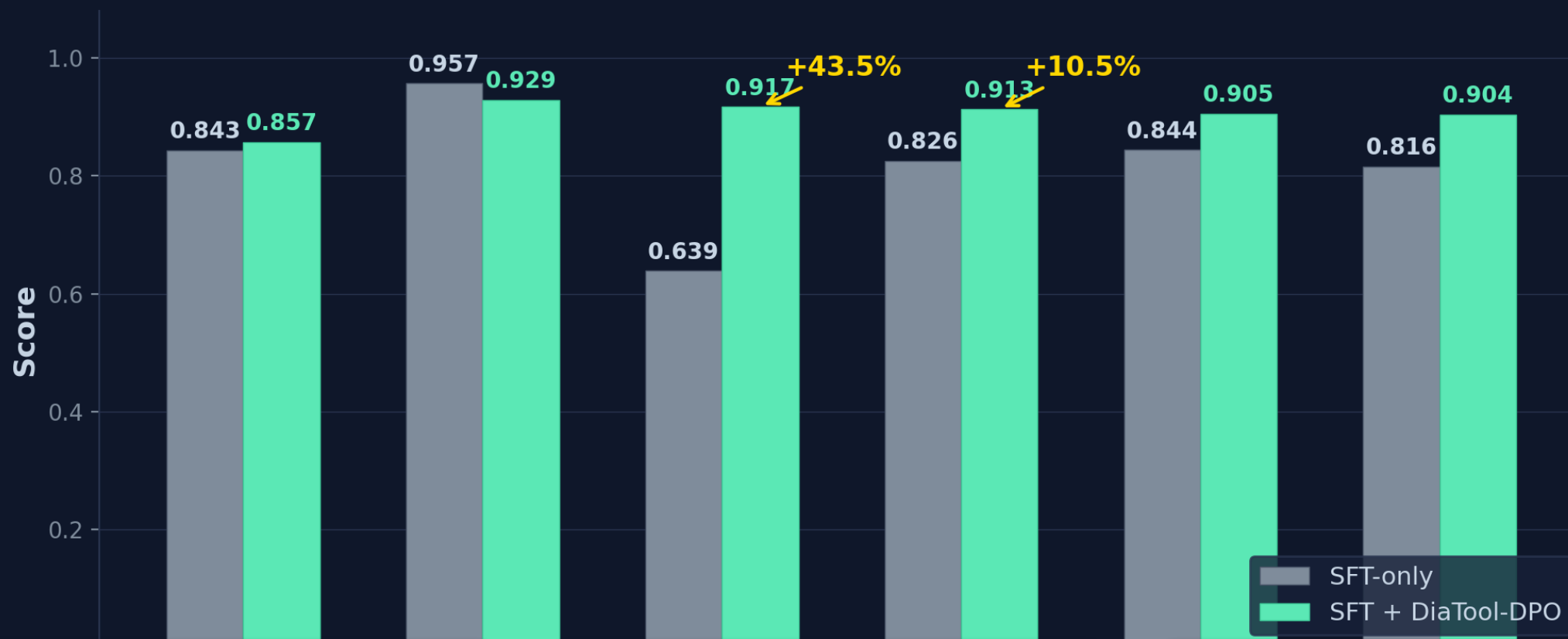
FunctionChat-Bench evaluation across 4 metrics · Bold = best per model · Green = best overall Macro Avg

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

Key: DPO alone is insufficient — it complements SFT. SFT + DiaTool-DPO achieves best balanced performance (Macro Avg) across all models.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance

Impact of adding DiaTool-DPO to SFT on LLaMA-3-8B · FunctionChat-Bench evaluation



Slot: 0.639 → 0.917

+43.5% improvement

Relevance: 0.826 → 0.913

+10.5% improvement

Near GPT-4o Performance

94.8% info gathering, 91% rejection

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1

MDP Formulation Enables Systematic Pairing

5 internal states and 3 query types allow automatic generation of chosen/rejected trajectory pairs — no human annotation required.

2

DPO Complements SFT, Does Not Replace It

DiaTool-DPO alone performs poorly. Combined with SFT, it achieves the best balanced Macro Average across all model sizes.

3

Largest Gains in Slot-Filling & Tool Rejection

LLaMA-3-8B: +43.5% Slot accuracy, +10.5% Relevance. These are the exact failure modes SFT struggles with most.

4

Language-Agnostic & Production-Ready

Primary experiments in Korean generalize to English. Opens a path to GPT-4o-level conversational tool use on smaller, deployable open-source models.

5

Automatic Dataset Construction at Scale

Paired trajectories generated automatically from the MDP structure — eliminates the human labeling bottleneck for preference optimization in dialogue systems.